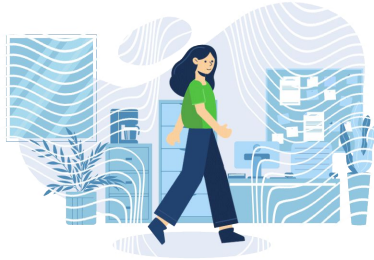


WiFi Sensing



WiFi Sensing

WiFi Access
Point



WiFi Receiving
Device



What is CSI?



Channel State Information (CSI) of WiFi Signals

- Utilized primarily for advanced techniques that enhance wireless communication performance and reliability.
- Advance techniques: beamforming, adaptive modulation and coding scheme, spatial diversity and so on.
- CSI provides information about the characteristics of the wireless signals.

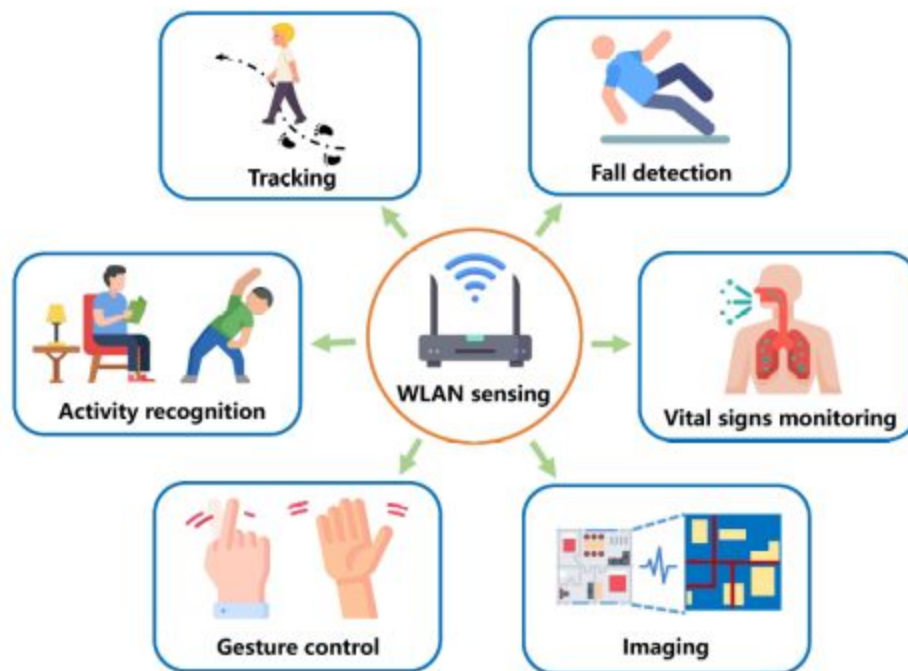
What is CSI?



- Changes in signal amplitude and phase as they are affected by the environment and objects.
- CSI helps the WiFi to understand these changes in the air.
- It helps to sense the environment.

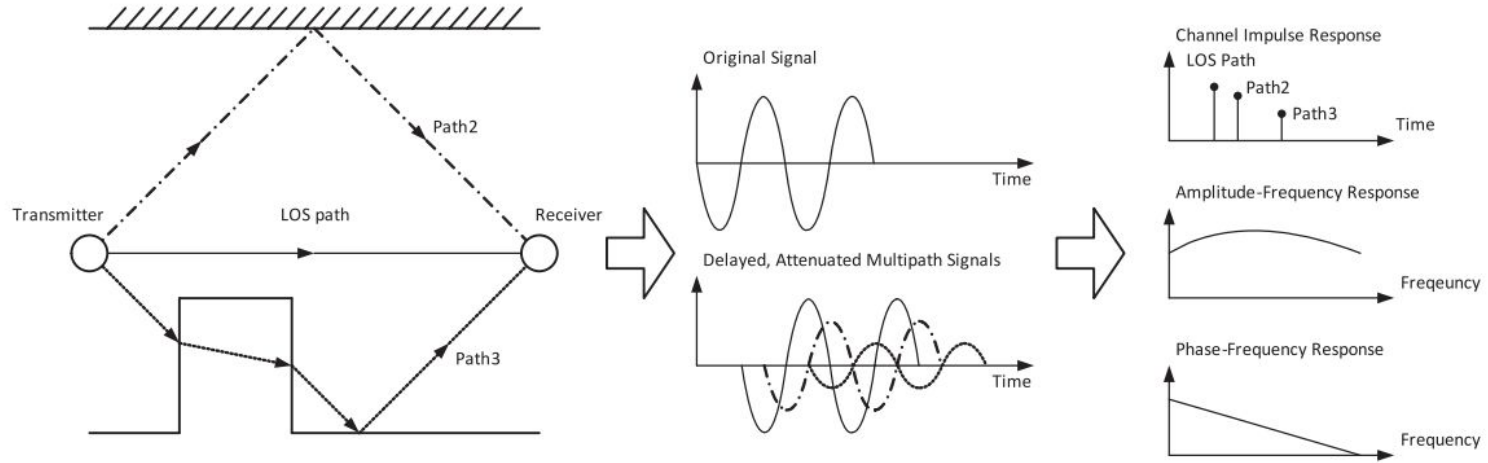
WiFi Sensing using CSI

Sensing

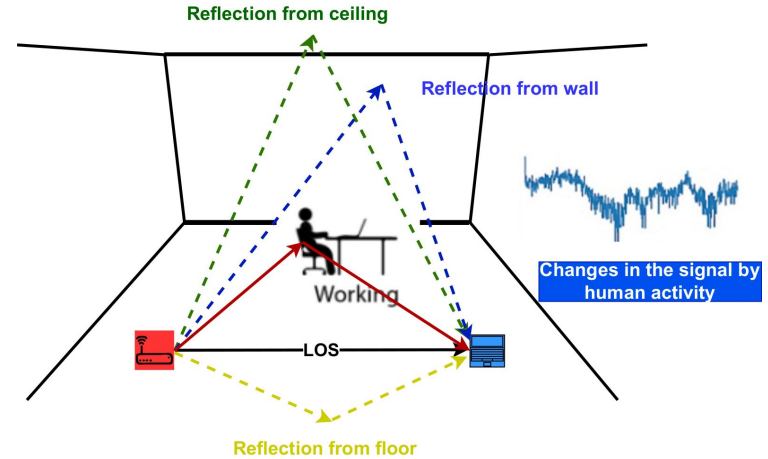
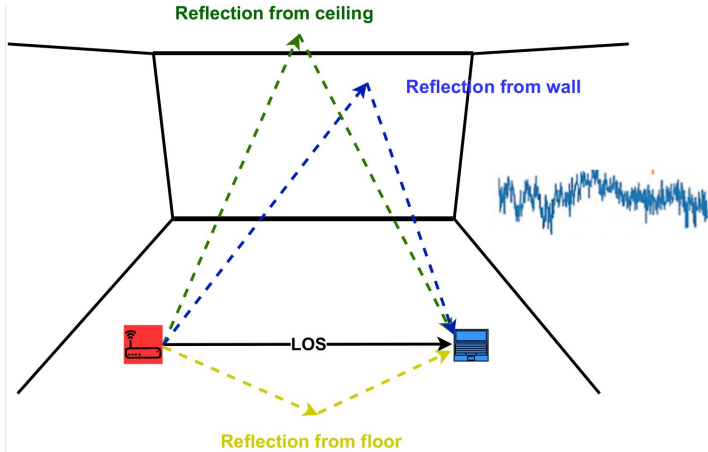


Support device-free target sensing

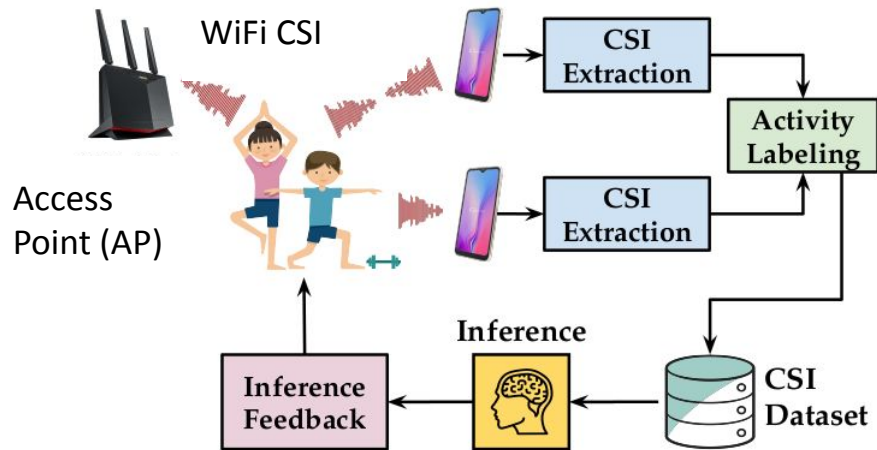
Multipath Propagation



What is WiFi Sensing using CSI ?

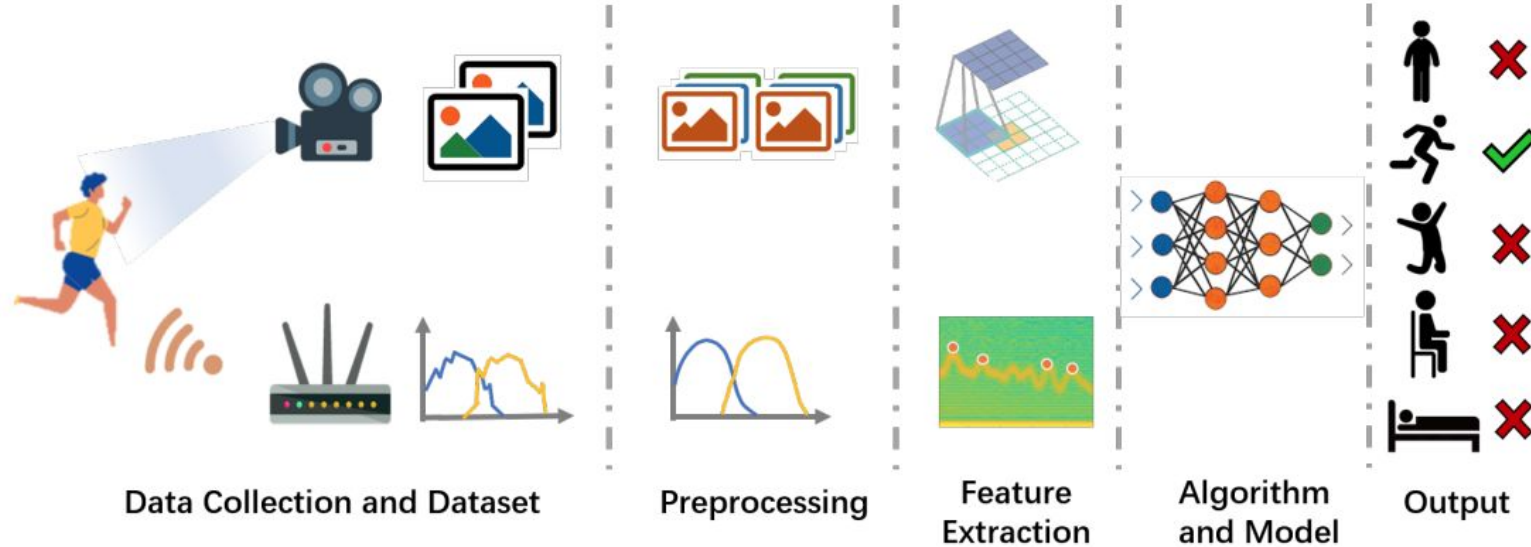


Applications



- Human Activity Recognition (HAR)
- Human Gesture Recognition (HGR)
- Person detection and identification
- Respiration monitoring
- More..

WiFi Sensing Steps



WiFi Sensing: steps

